

```
!pip install plotly
```

Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (5.24.1)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from plotly) (26.1)

Import Libraries

```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
```

Load Dataset

```
df = pd.read_csv('/content/Sample - Superstore.csv', encoding='latin1')

df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Product ID
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-BO-10001798
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-CH-10000454
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036	West	OFF-LA-10000240
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	FUR-TA-10000577
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	OFF-ST-10000760

5 rows × 21 columns

Insights

```
print(df.info())

print(df.columns)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Row ID              9994 non-null  int64
1   Order ID            9994 non-null  object
2   Order Date          9994 non-null  object
3   Ship Date           9994 non-null  object
4   Ship Mode           9994 non-null  object
5   Customer ID         9994 non-null  object
6   Customer Name       9994 non-null  object
7   Segment             9994 non-null  object
8   Country             9994 non-null  object
9   City                9994 non-null  object
10  State               9994 non-null  object
11  Postal Code         9994 non-null  int64
12  Region              9994 non-null  object
```

```
13 Product ID      9994 non-null object
14 Category        9994 non-null object
15 Sub-Category    9994 non-null object
16 Product Name    9994 non-null object
17 Sales           9994 non-null float64
18 Quantity        9994 non-null int64
19 Discount        9994 non-null float64
20 Profit          9994 non-null float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
None
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
      'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
      'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
      'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],
      dtype='object')
```

Total Sales KPI Analysis

```
total_sales = df['Sales'].sum()

print("Total Sales:", total_sales)
```

```
Total Sales: 2297200.8603000003
```

Total Profit KPI

```
total_profit = df['Profit'].sum()

print("Total Profit:", total_profit)
```

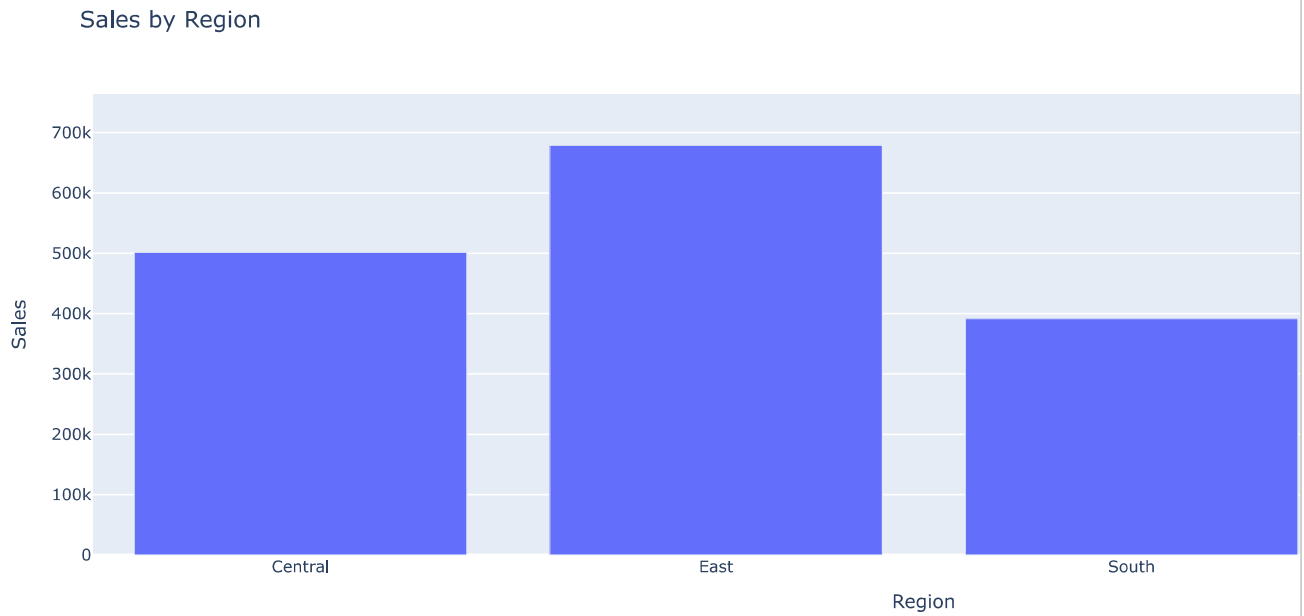
```
Total Profit: 286397.0217
```

Sales By Region

```
region_sales = df.groupby('Region')['Sales'].sum().reset_index()

fig = px.bar(
    region_sales,
    x='Region',
    y='Sales',
    title='Sales by Region'
)

fig.show()
```



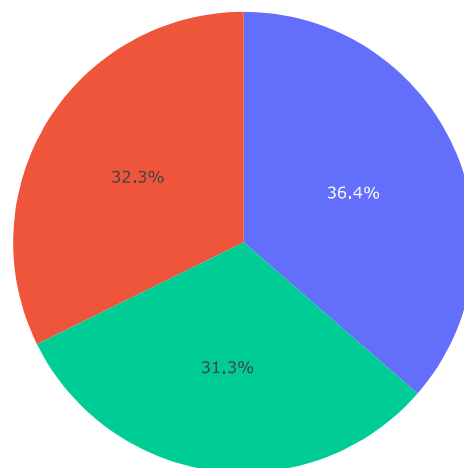
Category Sales PIE Chart

```
category_sales = df.groupby('Category')['Sales'].sum().reset_index()

fig = px.pie(
    category_sales,
    names='Category',
    values='Sales',
    title='Sales by Category'
)

fig.show()
```

Sales by Category



Monthly Sales Trend

```
# Convert date column

df['Order Date'] = pd.to_datetime(df['Order Date'])

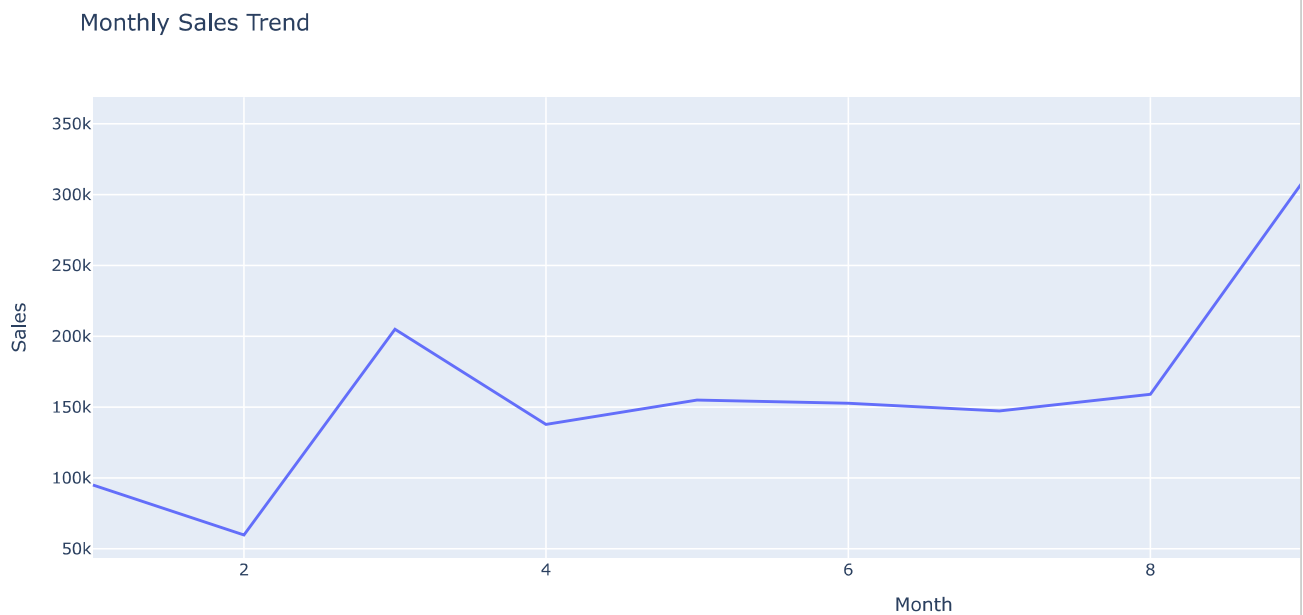
# Extract month

df['Month'] = df['Order Date'].dt.month

monthly_sales = df.groupby('Month')['Sales'].sum().reset_index()

fig = px.line(
    monthly_sales,
    x='Month',
    y='Sales',
    title='Monthly Sales Trend'
)

fig.show()
```



Top Products sold

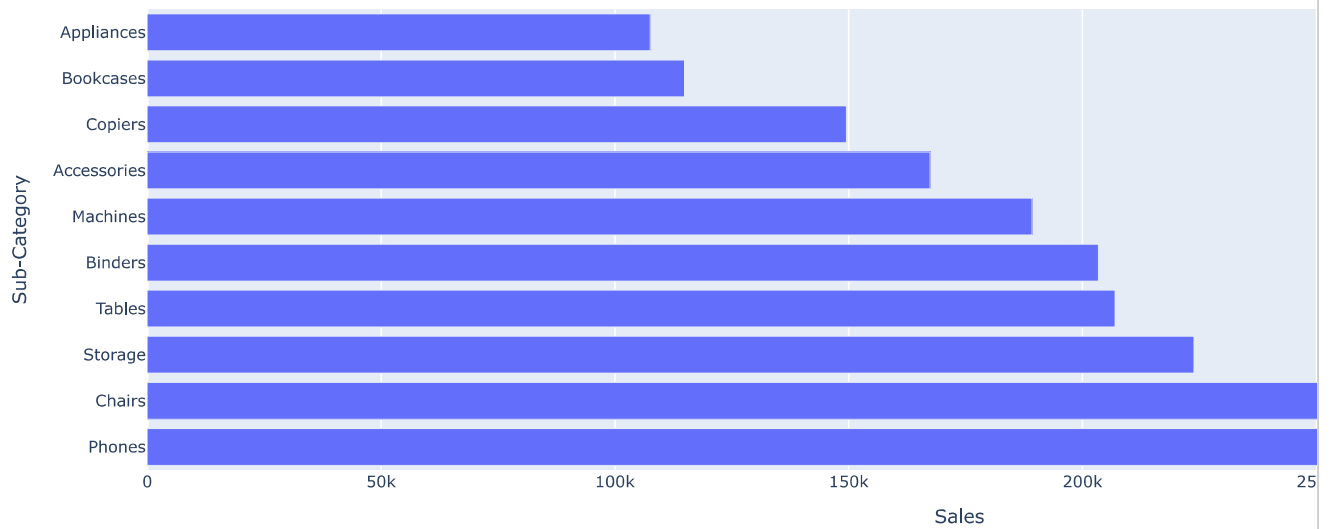
```
top_products = df.groupby('Sub-Category')['Sales'].sum().reset_index()

top_products = top_products.sort_values(
    by='Sales',
    ascending=False
).head(10)

fig = px.bar(
    top_products,
    x='Sales',
    y='Sub-Category',
    orientation='h',
    title='Top 10 Products'
)

fig.show()
```

Top 10 Products



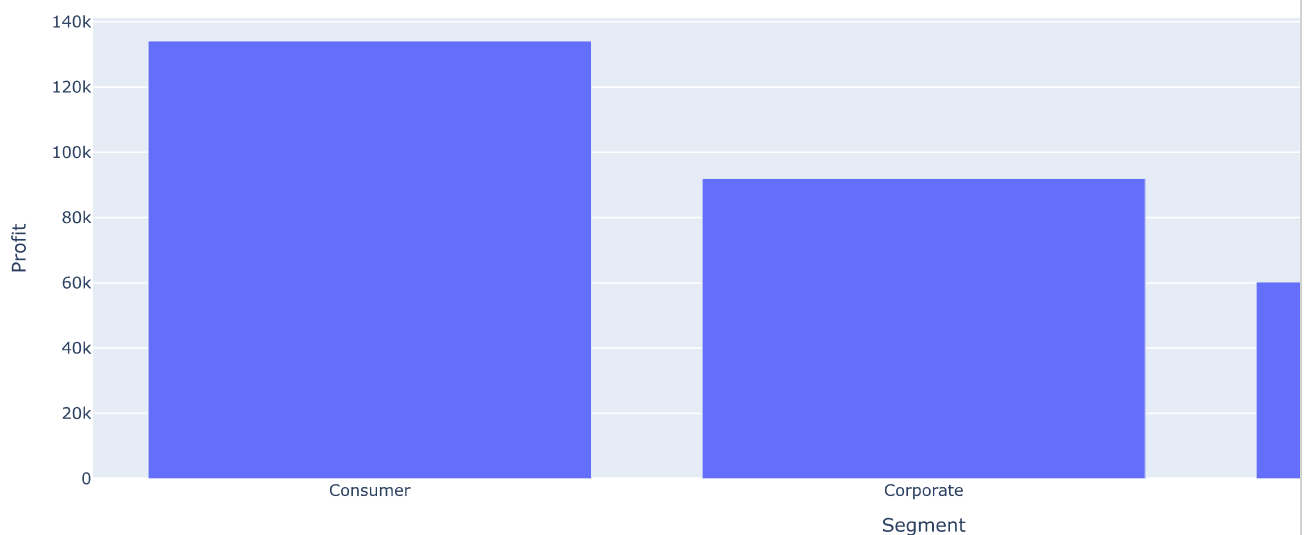
Profit By Segment

```
segment_profit = df.groupby('Segment')['Profit'].sum().reset_index()

fig = px.bar(
    segment_profit,
    x='Segment',
    y='Profit',
    title='Profit by Segment'
)

fig.show()
```

Profit by Segment



Sales vs Profit (Scatter plot)

```
fig = px.scatter(  
    df,  
    x='Sales',  
    y='Profit',  
    color='Category',  
    title='Sales vs Profit'  
)  
  
fig.show()
```

